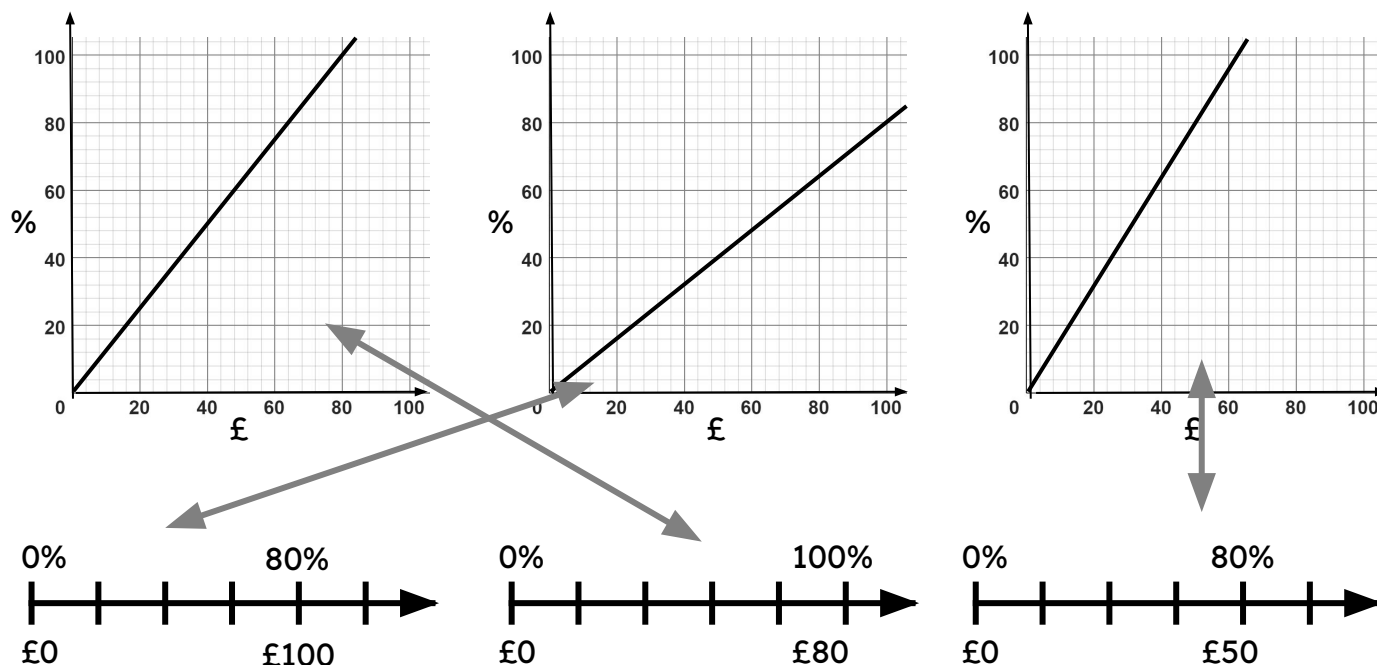




Multiplicative relationships presented graphically

Task A: Percentages of amounts graphically

1) Match the double number line with the corresponding graph.



2) A class of students took a test. The grades available were A, B, C or D.

The graph helps the teacher convert a score out of 65 to a grade.

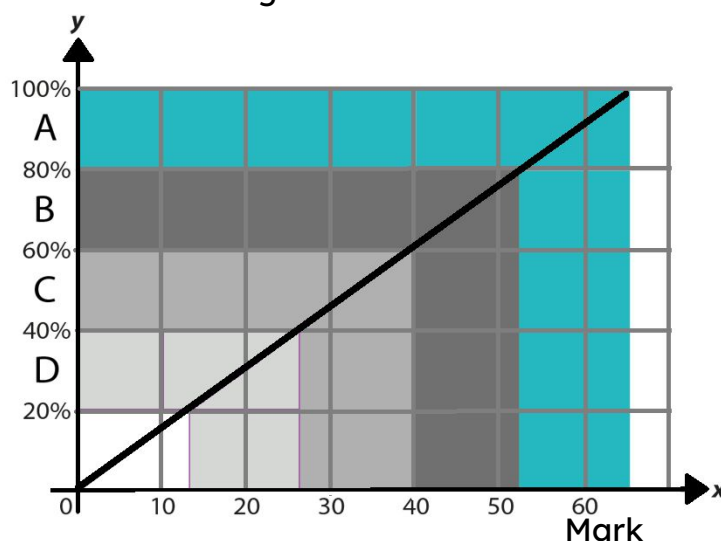
Jun scored 35 marks on his assessment.

a) What grade did Jun get?

35 marks is a grade C.

b) Estimate his percentage score.

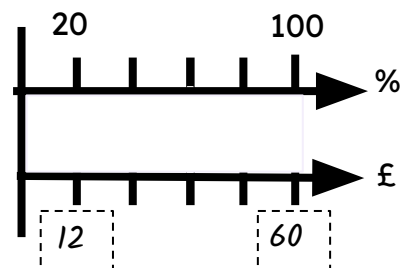
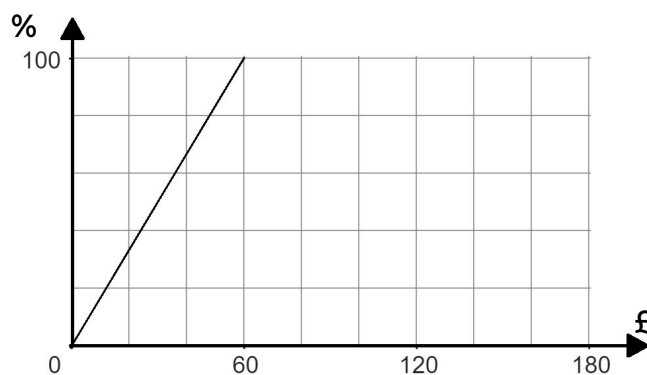
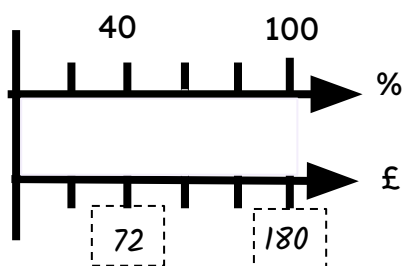
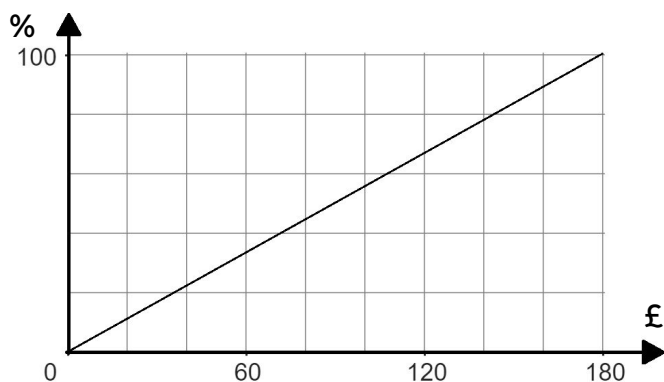
Jun scored 54% (an estimate between 50% and 60% is reasonable).



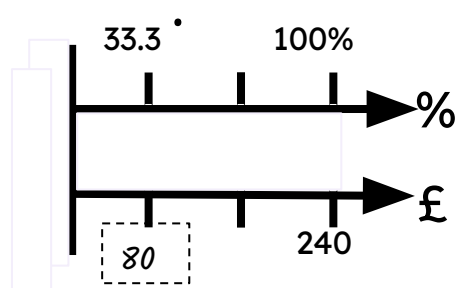
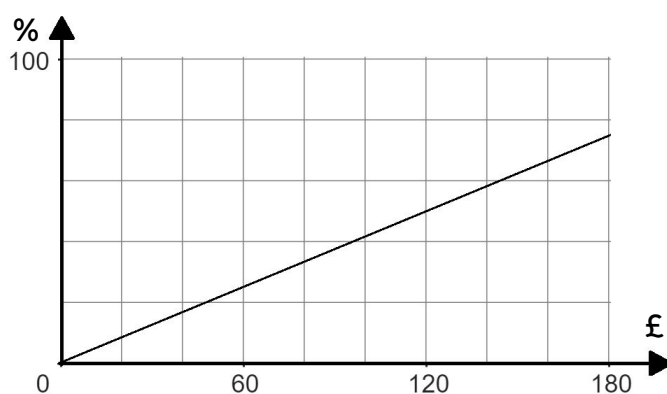
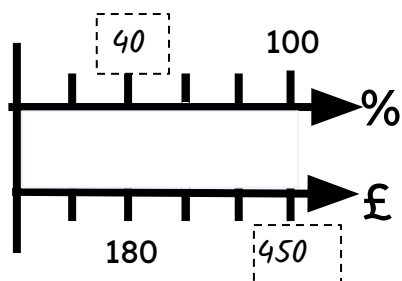
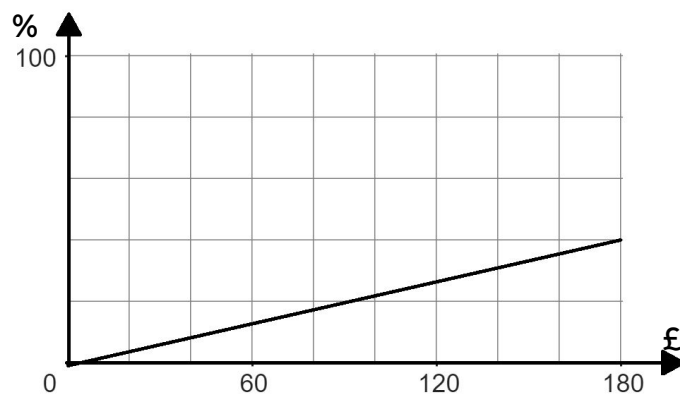
c) Estimate the minimum mark you need to get a grade. What calculation could you carry out to be more accurate?

13 is the minimum mark needed for a grade. The calculation is 65×0.2 (or equivalent).

3a) Fill in the missing information given the graph and double number line.



3b) Fill in the missing information given the graph and double number line.



Task B: Other representations of proportional relationships

1) Write what you think are the advantages and disadvantages of using a graph, double number line and ratio table when trying to find a percentage of an amount.

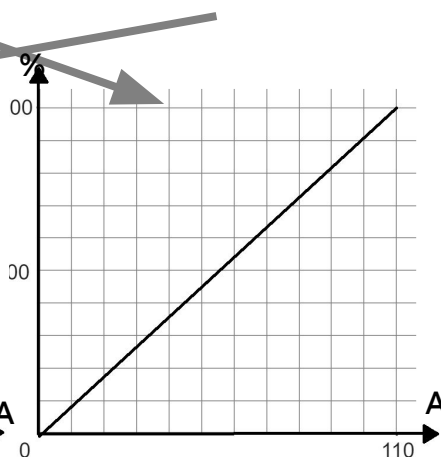
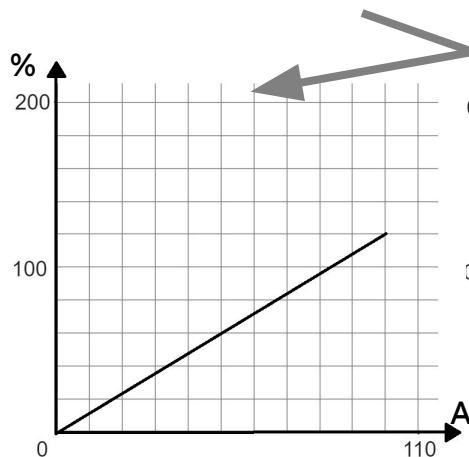
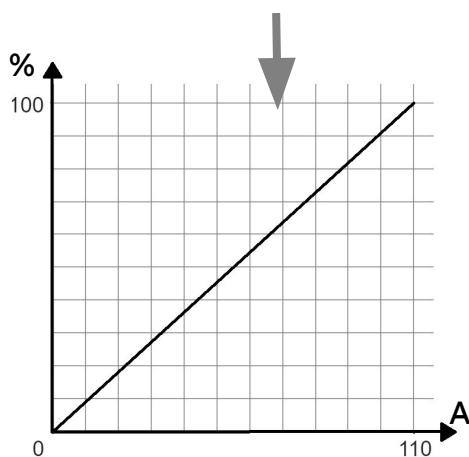
	Graph	Double number line	Ratio table
Advantages	Good visual representation	Good visual representation	Concise
Disadvantages	Takes time to draw Answers are dependent on the accuracy of the scale	Takes time to draw as it is scaled	Doesn't show visual representation

2) Match the ratio table with the correct graph and work out the missing amount using the ratio table.

Amount	%
110	100
60.5	55

Amount	%
55	100
30.25	55

Amount	%
100	120
30	36



Amount	%
110	100
55	50%
5.5	5%
60.5	55%

Amount	%
55	100
27.5	50%
2.75	5%
30.25	55%

Amount	%
100	120
10	12%
30	36%

